ANALYSTLAB AFRICA

Machine Learning Internship Programme

Data Preprocessing Report

From Raw Records to a Machine-Learning-Ready Dataset

Week 2 — Data Preprocessing & Feature Engineering for Machine Learning

Programme	Machine Learning Internship
Project	Data Preprocessing & Feature Engineering for Machine Learning
Deliverable	Data Preprocessing Report
Dataset	Telco Customer Churn (7,043 rows × 21 columns)
Version	1.0

1. Dataset Inspection

The Telco Customer Churn dataset (Telco-Customer-Churn.csv) was loaded and inspected before any transformation.

Check	Result
Shape	7,043 rows × 21 columns
Duplicate rows	0
Missing values (raw isnull() scan)	0 reported — see Section 2 for the hidden case found
Categorical columns	18 (including customerID and TotalCharges, mistakenly typed as object)
Numerical columns	3 (SeniorCitizen, tenure, MonthlyCharges)

2. Data Quality Issues Identified

2.1 TotalCharges stored as text

Data.info() shows TotalCharges as an object column, despite holding numeric amounts. Investigation (Data[Data['tenure'] == 0]) revealed the cause: 11 customers with tenure == 0 have an empty string instead of a numeric value in TotalCharges — pandas defaults the entire column to text when it encounters a non-numeric entry.

```
Data["TotalCharges"] = pd.to_numeric(Data["TotalCharges"], errors="coerce")
```

This coercion converts the column to float64 and turns the 11 empty strings into NaN — an initially hidden missing-value issue that the standard isnull().sum() scan on the raw column did not catch, because the column was still text at that point.

2.2 Target variable typed as text

Churn is stored as "Yes"/"No" text. Since it is the prediction target, it must be numeric for any downstream classifier.

```
Data.Churn = Data["Churn"].map({"No": 0, "Yes": 1})
```

2.3 Category consistency check

Every column's unique() and value_counts() were inspected to rule out inconsistent labels (typos, mixed casing, duplicated meanings). No inconsistencies were found — categories such as OnlineSecurity or InternetService use clean, consistent labels (e.g. "Yes", "No", "No internet service").

3. Feature Engineering

3.1 tenure_group

tenure was binned into six 12-month intervals to help a model capture the non-linear, gradually declining relationship between customer tenure and churn probability.

```
bins = [-1, 12, 24, 36, 48, 60, 72]
labels = ['0-12', '13-24', '25-36', '37-48', '49-60', '61-72']
Data["tenure_group"] = pd.cut(Data['tenure'], bins=bins, labels=labels)
```

3.2 is_new_customer

A binary risk flag isolates the 0–12 month tenure band, which the exploratory count/churn-rate plots identified as the highest-risk segment (~47% churn rate in that group).

```
Data["is_new_customer"] = (Data['tenure'] <= 12).astype(int)
```

Why engineer both a bin and a flag?

tenure_group preserves the full ordinal structure for the model to learn a gradual risk gradient, while is_new_customer gives an explicit, always-available binary signal for the single highest-risk segment — useful even for simpler models that cannot exploit ordinal category structure on their own.

4. Outlier Detection & Treatment

Box plots and histograms of tenure, MonthlyCharges, and TotalCharges showed no visually extreme values. This was then verified quantitatively with the IQR method (1.5×IQR bounds) applied to all three numerical features:

```
for col in num_feature:
    q_1 =
Data[col].quantile(0.25)
    q_3 =
Data[col].quantile(0.75)
    IQR = q_3 - q_1
    lowest_range = q_1 - 1.5 *
IQR
    biggest_range = q_3 + 1.5 *
IQR
    Data = Data[(Data[col] >=
lowest_range) & (Data[col] <=
biggest_range)]
```

Metric	Value
Rows before IQR filtering	7,043
Rows after IQR filtering	7,032

Rows removed	11 (0.16% of the dataset)

The 11 rows removed are exactly the tenure == 0 customers with missing TotalCharges identified in Section 2.1: NaN values automatically fail the IQR boundary comparison. This step therefore resolved that missing-value issue as a side effect of outlier filtering, confirming the box-plot assessment that the dataset contains no meaningful outliers beyond these incomplete records.

5. Feature Selection

5.1 Correlation analysis (numerical features)

A correlation heatmap of the numerical features (SeniorCitizen, tenure, MonthlyCharges, TotalCharges, Churn) showed strong correlation between TotalCharges and both tenure and MonthlyCharges — expected, since TotalCharges is mathematically derived from them (tenure × average monthly billing).

Decision: TotalCharges was dropped to remove multicollinearity and redundant information, since tenure and MonthlyCharges already capture the same signal independently.

5.2 Statistical testing (categorical features)

A correlation heatmap cannot evaluate categorical variables. gender, PhoneService, and MultipleLines were therefore tested against Churn using a Chi-square test of independence:

```
from scipy.stats import
chi2_contingency
for col in ['gender',
'PhoneService',
'MultipleLines']:
    contingency =
pd.crosstab(Data[col],
Data['Churn'])
    chi2, p, dof, expected =
chi2_contingency(contingency)
```

Feature	Chi^2	p-value	Conclusion
gender	0.48	0.4905	No significant association
PhoneService	0.87	0.3499	No significant association
MultipleLines	11.27	0.0036	Statistically significant

MultipleLines returned a significant p-value, which is not sufficient on its own to justify removal. Cramer's V was computed to measure the practical strength of the association, independent of the large sample size (n = 7,043):

```
def cramers_v(chi2, contingency_table):
    n = contingency_table.sum().sum()
```

```
    r, k = contingency_table.shape
    return np.sqrt(chi2 / (n * (min(r, k) - 1)))

# Cramer's V for MultipleLines: 0.0400
```

Interpretation

Cramer's V = 0.04 falls below the 0.1 threshold generally considered "negligible". MultipleLines is statistically associated with Churn but the effect size is not practically meaningful — confirming it can be removed without meaningful loss of predictive information.

Final decision: gender, PhoneService, MultipleLines, customerID (a unique identifier with no predictive value), and TotalCharges were dropped from the training set.

5.3 Independence check (tenure vs. MonthlyCharges)

A scatter plot of MonthlyCharges against tenure showed a fully dispersed cloud with no visible pattern, confirming the two retained numerical features are independent and both worth keeping for modeling.

5.4 Variance Threshold (post-encoding)

After One-Hot Encoding (Section 6), a Variance Threshold check was run on the resulting binary columns plus is_new_customer, excluding the already-scaled tenure and MonthlyCharges and the target Churn:

```
ohe_columns = [c for c in transformed_data.columns
               if c not in ["tenure", "MonthlyCharges", "Churn", "tenure_group"]]
selector = VarianceThreshold(threshold=0.01)
selector.fit(transformed_data[ohe_columns])

# Columns removed due to low variance: []
```

No columns were removed. Every encoded feature — including is_new_customer, whose high-risk segment represents roughly 30% of customers — carries sufficient variance to remain potentially informative.

6. Feature Encoding & Scaling

A single ColumnTransformer applied three encoding strategies in one coordinated step, matched to each feature's nature:

Feature type	Technique	Applied to
Numerical, continuous	StandardScaler	tenure, MonthlyCharges
Categorical, nominal	One-Hot Encoding	All remaining object-type columns
Categorical, ordinal	Ordinal Encoding	tenure_group
Already binary numeric	Passthrough (unchanged)	SeniorCitizen, is_new_customer, Churn

StandardScaler was chosen over MinMax or Robust scaling because the IQR analysis (Section 4) confirmed no significant outliers in the numerical features — the main scenario where a robust scaler would be preferred. One-Hot Encoding was used rather than Label Encoding for nominal categories, since these categories have no inherent order and label encoding would incorrectly imply one. tenure_group, by contrast, is genuinely ordinal (0-12 < 13-24 < ... < 61-72), so Ordinal Encoding was used specifically for that column to preserve its ranking.

7. Final Machine-Learning-Ready Dataset

Property	Value
Final shape	7,032 rows × 40 columns
Rows removed vs. raw dataset	11 (outlier / missing-value treatment)
Dropped columns	customerID, gender, PhoneService, MultipleLines, TotalCharges
Engineered columns added	tenure_group, is_new_customer
Encoding applied	One-Hot (nominal) + Ordinal (tenure_group)
Scaling applied	StandardScaler (tenure, MonthlyCharges)
Export file	Telco-Customer-Processed-Churn.csv

Readiness statement

The exported dataset contains no identifier columns, no missing values, no duplicate rows, no unresolved outliers, and every feature is in a numeric, model-consumable form. It is ready for train/test splitting and model development in the next phase of the internship programme.